

KERALA AGRICULTURAL UNIVERSITY

[Home](#)[Institution](#)[Education](#)[Research](#)[Extension](#)[News](#)[Contact Us](#)[Back to Contents](#)

VEGETABLES

Under this Chapter, package of practices for amaranth, okra, cucurbitaceous vegetables, solanaceous vegetables, cool season vegetables and minor vegetables are furnished. A separate table showing the waiting period for insecticides on vegetables has been included. Control of pests of vegetables with the help of non-chemical insecticides and tips for vegetable seed production are also given under separate titles. Certain recommendations given under the non-chemical control of pests have not been included under the respective crops to avoid repetition.

AMARANTH (*Amaranthus* spp.)

Amaranth is the most popular leafy vegetable of Kerala. It can be grown throughout the year. Avoid sowing or planting of red leaved varieties during periods of heavy rain.

Varieties

Red: Kannara Local, Arun and Krishnasree.

Green: Co-1, Co-2, Co-3, Mohini and Renusree

Note : Kannara Local is a season bound variety, which comes to flowering in November-December.

Seed rate: 1.5 to 2.0 kg ha⁻¹

Preparation of land

Prepare the land by ploughing or digging followed by levelling. Then shallow trenches of width 30-35 cm are made 30 cm apart. Well rotten FYM is mixed with soil in the trenches. Transplant 20-30 day old seedlings in the shallow trenches at a distance of 20 cm in two rows. During rainy season planting shall be done on raised beds.

Manuring

Apply 50 tonnes of FYM per ha as basal dose before planting. After preparing trenches, apply N:P₂O₅:K₂O @ 50:50:50 kg ha⁻¹. Another 50 kg of N can be applied at regular intervals as top dressing. Spraying 1 per cent urea immediately after each harvest will increase the yield.

Plant protection

As far as possible, avoid use of insecticides or fungicides. In severe cases of leaf webber attack, spray malathion 0.1 per cent or dust malathion 10 per cent DP.

[Top](#) [Back to Contents](#)

OKRA (*Abelmoschus esculentus*)

The three main planting seasons for Okra are February-March, June-July and October-November.

Varieties

Green / light green fruited: Pusa Sawani, Pusa Makhmali, IARI Selection 2, Kiran, Salkeerthi.

Red fruited: Co-1, Aruna

Yellow vein mosaic resistant/tolerant: Arka Anamika, Arka Abhay, Susthira, Anjitha, Manjima (all green fruited).

Seed rate

The seed rate is 8.5 kg ha⁻¹ for the summer crop sown in February-March and 7 kg ha⁻¹ for kharif crop.

Storage of seeds

Packing of okra seeds in polythene cover (700 gauge) increases the storage life upto 7 months.

Planting

For kharif crop, sow the seeds at a spacing of 60 cm between rows and 45 cm between plants. For the summer crop, soak the seeds in water for 24 hours before sowing and give a spacing of 60 cm x 30 cm.

Manuring

Apply FYM or compost as basal dose @ 12 t ha⁻¹. At the time of sowing apply N, P₂O₅ and K₂O @ 55, 35 and 70 kg ha⁻¹. Another 55 kg N ha⁻¹ may be applied one month after sowing.

Note: For reclaimed soils of Kuttanad, a fertilizer dose of N:P₂O₅:K₂O 75:5:15 kg ha⁻¹ is recommended.

Aftercultivation

Give pre-sowing irrigation, if soil is not moist enough. During summer, irrigate at intervals of 2 to 3 days. Conduct weeding regularly and earth up rows during rainy season.

Plant protection

The important pests are jassids, fruit and shoot borer and root knot nematode.

Against jassids, use quinalphos 0.05 per cent as foliar sprays. For controlling fruit and shoot borers, remove all drooping shoots and damaged fruits. Spray carbaryl 0.15 per cent at intervals of 15 to 20 days. For controlling aphids, apply dimethoate 0.05 per cent.

For the control of nematodes, apply sawdust or paddy husk at 500 g/plant or neem leaves or *Eupatorium* leaves at 250 g/plant in basins one week prior to planting and water daily. The effect of this treatment persists up to 75 days after sowing in summer season.

For managing root knot nematode, seed treatment with *Bacillus macerans* @ 3 per cent w/w. (2.5 kg ha⁻¹) and in heavily infested area seed treatment with *B. macerans* @ 3 per cent w/w. and drenching with *B. macerans* @ 3 per cent solution 30 days after sowing can be recommended.

In general, insecticides of plant origin may be used, as far as possible.

Yellow vein mosaic

This is a common disease in okra, which shows vein clearing and vein chlorosis of leaves. The yellow network of veins is very conspicuous and veins and veinlets are thickened. Fruits become small and yellowish green in colour. White fly (*Bemisia tabaci*) and leaf hopper (*Amrasca biguttula biguttula*) are vectors of this virus. Hence, their control is very important. Use of resistant varieties like Arka Anamika, Arka Abhay and Susthira, and destruction of host weeds (*Croton sparsiflora* and *Ageratum* sp.) are also effective.

[Top](#) [Back to Contents](#)

CUCURBITACEOUS VEGETABLES**BITTER GOURD (*Momordica charantia*)**

Bitter gourd is an important cucurbitaceous vegetable of Kerala.

Season

January-March and September-December are the ideal seasons. For the rainfed crop, sowing can be started after the receipt of first few showers during May-June.

Varieties

Priya, Preethi, Priyanka and Arka Harit are high yielding varieties. Priyanka is recommended for acid alluvial soils of Kerala.

Seed rate: 5.0 _ 6.0 kg ha⁻¹

Spacing: 2.0 m x 2.0 m

Sowing

Pits of 60 cm diameter and 30-45 cm depth are taken. Well rotten FYM and fertilizers are mixed with topsoil in the pit and seeds are sown @ 4-5 per pit. Unhealthy plants are removed after two weeks and only 3 plants are retained per pit.

Manuring

Apply FYM @ 20-25 t ha⁻¹ as basal dose along with half dose of N (35 kg) and full doses of P₂O₅ (25 kg) and K₂O (25 kg). The remaining dose of N (35 kg) can be applied in several split doses at fortnightly intervals.

Aftercultivation

During the initial stages of growth, irrigate at 3-4 days interval and alternate days during flowering/fruiting. Irrigation at 15 mm CPE (approximately at 3 days interval for sandy loam soils) is more economical than irrigating once in two days especially during summer months for water economy. Erect pandals when the plants start vining. Conduct weeding and raking of the soil at the time of fertilizer application. Earthing up is done during rainy season.

Pests**Fruit fly (*Bacterocera cucurbitae*)**

Apply carbaryl 10 per cent DP in pit before sowing of seeds to destroy the pupae. In homestead gardens the fruits may be covered with polythene, cloth or paper bags to ensure mechanical protection. In large gardens apply carbaryl 0.2 per cent or malathion 0.15 per cent suspension containing sugar or jaggery at 10 g/l at fortnightly intervals at flowering and fruit initiation. Spray as coarse droplets on the ventral surface of leaves. Remove and destroy affected and decayed fruits. It can also be effectively controlled by the use of banana fruit traps coupled with the removal and destruction of infested fruits. It is more efficient than two sprayings with insecticides. Traps are to be set at a distance of 2 m after a border row and they may be replenished after 7 to 9 days. Start bait trapping just before flowering.

Epilachna beetle

Remove and destroy egg masses, grubs and adults occurring on leaves. Spray carbaryl 0.2 per cent.

Red pumpkin beetle

Adult beetle eats the leaves, makes hole on foliage and causes damage on roots and leaves. Incorporate carbaryl 10 per cent DP in pits before sowing the seeds to destroy grubs and pupae.

Plant lice

Apply 1.5 per cent fish oil soap. First dissolve soap in hot water and then make up the volume. Alternatively apply dimethoate 0.05 per cent.

Diseases**Downy mildew**

It is severe during rainy season. This can be checked by spraying mancozeb 0.2 per cent (waiting period of mancozeb is three days).

Powdery mildew

Can be controlled by spraying dinocap 0.05 per cent.

Mosaic

Control the vectors by spraying dimethoate 0.05 per cent. Uprooting and destruction of affected plants and collateral hosts should be done.

Harvesting can be done only after 10 days (at least) of insecticide / fungicide application. The fruits should be washed thoroughly in water before cooking.

[Top](#) [Back to Contents](#)

SNAKE GOURD (*Trichosanthes cucumerina*)

Snake gourd is a common cucurbitaceous vegetable of Kerala.

Season

January-March and September _ December are the main growing seasons.

Varieties : Kaumudi, Baby, TA-19 and Manusree

Seed rate: 3.0 _ 4.0 kg ha⁻¹

Spacing : 2.0 m x 2.0 m

Sowing

Pits of 60 cm diameter and 30-45 cm depth are taken. Well rotten FYM and fertilizers are mixed with top soil in the pit and seeds are sown @ 4-5 per pit. After two weeks remove the unhealthy plants retaining three plants per pit.

Manuring

Apply FYM @ 20-25 t ha⁻¹ as basal dose along with half dose of N (35 kg) and full dose of P₂O₅ (25 kg) and K₂O (25 kg). The remaining dose of N (35 kg) is applied in several split doses at fortnightly intervals.

Aftercultivation

During the initial stages of growth irrigate at an interval of 3-4 days. Irrigate on alternate days during flowering and fruiting periods.

Erect pandals for trailing snake gourd. Conduct weeding and raking of the soil at the time of fertilizer application.

Pests

The important pests attacking snake gourd are fruit flies, epilachna beetle and red pumpkin beetle. Control measures recommended for bitter gourd can be followed in this case also.

Diseases

The important diseases are downy mildew and mosaic. The control measures as stated under bitter gourd can be adopted. Harvesting can be done only after 10 days (at least) of insecticide / fungicide application. The fruits should be washed thoroughly in water before cooking.

[Top](#) [Back to Contents](#)

**CUCUMBER (*Cucumis sativus*) &
ORIENTAL PICKLING MELON (*Cucumis melo* var. *conomon*)**

Cucumber and oriental pickling melon are important cucurbits grown in Kerala. Cucumber is mainly used as a salad crop whereas oriental pickling melon is largely used after cooking.

Season

The ideal seasons are January-March and September-December.

Varieties

Mudicode, Arunima and Saubhagya are high yielding varieties of oriental pickling melon. Cucumber varieties such as Pusa Sheethal, Poinsette, Poona Khira are exclusively used for salad purpose.

Seed rate: 0.5 _ 0.75 kg ha⁻¹

Spacing: 2.0 m x 1.5 m

Preparation of land

Pits of 60 cm diameter and 30-45 cm depth are taken. Well rotten FYM and fertilizers are mixed with topsoil in the pit and four or five seeds are sown in a pit. Remove unhealthy plants after two weeks and retain three plants per pit.

Manuring

Apply FYM @ 20-25 t ha⁻¹ as basal dose along with half dose of N (35 kg) and full dose of P₂O₅ (25 kg) and K₂O (25 kg ha⁻¹). The remaining dose of N (35 kg) can be applied in two equal split doses at the time of vining and at the time of full blooming. A fertilizer dose of 70:25:25 kg N:P₂O₅: K₂O/ ha in several splits is recommended in Onattukara region. The fertilizer dose per pit would be 28:10:10 g N:P₂O₅:K₂O.

Aftercultivation

During the initial stages of growth, irrigate at an interval of 3-4 days. Irrigate in alternate days, during flowering and fruiting. For trailing cucumber and melon, spread dried twigs on the ground. Conduct weeding and raking of the soil at the time of fertilizer application. Earthing up may be done during rainy season.

Pests

The important pests are epilachna beetle and red pumpkin beetle. They can be controlled by adopting the measures recommended for bitter gourd.

Diseases

The important diseases are downy mildew, powdery mildew and mosaic. The control measures as recommended for bitter gourd can be adopted.

Harvesting can be done only after 10 days (at least) of insecticide / fungicide application. The fruits should be washed thoroughly in water before cooking.

WATER MELON (*Citrullus lanatus*)

Season : The ideal season is Dec-April.

Varieties : Sugar Baby, Arka Jyothi

Seed rate : 1 to 1.5 kg ha⁻¹

Spacing : 3.0 m x 2.0 m

Preparation of land

Pits of 60 cm diameter and 30-45 cm depth are taken. Well rotten FYM and fertilizers are mixed with topsoil in the pit and four or five seeds are sown in a pit. Remove unhealthy plants after two weeks and retain two or three plants per pit.

Manuring

Apply FYM @ 20-25 t ha⁻¹ as basal dose along with half dose of N (35 kg) and full doses of P₂O₅ (25 kg) and K₂O (25 kg ha⁻¹). The remaining dose of N (35 kg) can be applied in two equal splits at the time of vining and full blooming.

Aftercultivation

During the initial stages of growth, irrigate at an interval of three or four days. Irrigate on alternate days during flowering and fruiting. When fruits mature, the frequency of irrigation may be reduced. For trailing the water melon, spread dried twigs on the ground. Conduct weeding and raking of the soil at the time of fertilizer application.

Pests

The important pests are red pumpkin beetle and epilachna beetle. They can be controlled by adopting measures suggested for bitter gourd.

Diseases

Downy mildew and powdery mildew are important diseases affecting water melon. Control them by adopting measures recommended for bitter gourd.

Harvesting can be done only after 10 days (at least) of insecticide / fungicide application. Only well ripe fruits should be harvested.

[Top](#) [Back to Contents](#)

BOTTLE GOURD (*Lagenaria siceraria*)

Season

Bottle gourd can be successfully grown during January-March and Sept-December. For the rainfed crop, sowing can also be started after the receipt of the first few showers during May-June.

Varieties

Pusa Summer Prolific Long, Arka Bahar

Seed rate: 3 _ 4 kg ha⁻¹

Preparation of land

Pits of 60 cm diameter and 30-45 cm depth are taken at 3 m x 3 m spacing for growing on pandals. For trailing on the ground, make pits at a spacing of 2 m in rows 3-4 m apart. Well rotten FYM and fertilizers are mixed with topsoil in the pit.

Sowing

Four or five seeds are sown per pit. Remove unhealthy plants after two weeks and retain three plants per pit.

Manuring

Apply FYM @ 20-25 t ha⁻¹ as basal dose along with half dose of N (35 kg) and full dose of P₂O₅ (25 kg ha⁻¹) and K₂O (25 kg ha⁻¹). The remaining dose of N (35 kg) can be applied in several split doses at fortnightly intervals.

Aftercultivation

During the initial stages of growth, irrigate at an interval of three or four days. Irrigate on alternate days during flowering and fruiting periods. Trailing can be done either on pandals or on the ground.

Conduct weeding and raking of the soil at the time of fertilizer application. Earthing up may be done during rainy season.

Pests

The important pests are epilachna beetle and red pumpkin beetle. They can be controlled by adopting measures recommended under bitter gourd.

Diseases

Downy mildew and powdery mildew are important diseases in bottle gourd. Refer control measures recommended for bitter gourd.

Harvesting can be done only after 10 days (at least) of insecticide / fungicide application. The fruits should be washed thoroughly in water before cooking.

[Top](#) [Back to Contents](#)

PUMPKIN (*Cucurbita moschata*)**Season**

Pumpkin can be successfully grown during January-March and September- December. For the rainfed crop, sowing can also be started after the receipt of the first few showers during May-June.

Varieties : Ambili, Suvarna, Saras and Sooraj

Seed rate: 1.0 to 1.5 kg ha⁻¹

Preparation of land

Pits of 60 cm diameter and 30-45 cm depth are taken at a spacing of 4.5 m x 2.0 m. Well rotten FYM and fertilizers are mixed with topsoil in the pit.

Sowing

Four or five seeds are sown per pit. Remove unhealthy plants after 2 weeks and retain three plants per pit.

Manuring

Apply FYM @ 20-25 t ha⁻¹ as basal dose along with half dose of N (35 kg) and full dose of P₂O₅ (25 kg) and K₂O (25 kg ha⁻¹). The remaining dose of N (35 kg) can be applied in two equal split doses at the time of vining and at the time of full blooming.

Aftercultivation

During the initial stages of growth, irrigate at an interval of three or four days. Irrigate on alternate days during flowering and fruiting periods. For trailing, spread dried twigs on the ground. Conduct weeding and raking of the soil at the time of fertilizer application. Earthing up may be done during rainy season.

Plant protection**Pests**

Fruit flies, epilachna beetle and red pumpkin beetle are important pests affecting pumpkin. They can be controlled as in case of bitter gourd.

Diseases

Downy mildew, powdery mildew and mosaic are important diseases affecting pumpkin.

Harvesting can be done only after 10 days (at least) of insecticide/fungicide application. The fruits should be washed thoroughly in water before cooking.

[Top](#) [Back to Contents](#)

ASH GOURD (*Benincasa hispida*)**Season**

Ash gourd can be successfully grown during January-March and September-December. For rainfed crop, sowing can also be started after the receipt of the first few showers during May-June.

Varieties: KAU Local, Indu

Seed rate: 0.75-1.0 kg ha⁻¹

Preparation of land

Pits of 60 cm diameter and 30-45 cm depth are taken at 4.5 m x 2.0 m spacing. Well rotten FYM and fertilizers are mixed with topsoil in the pit.

Sowing

Seeds are sown @ four or five per pit. Remove unhealthy plants after two weeks and retain two or three plants per pit.

Manuring

Apply FYM @ 20-25 t ha⁻¹ as basal dose along with half dose of N (35 kg) and full dose of P₂O₅ (25 kg) and K₂O (25 kg ha⁻¹). The remaining dose of N (35 kg) can be applied in two equal split doses at the time of vining and at the time of full blooming.

Aftercultivation

During the initial stages of growth, irrigate at an interval of three or four days. Irrigate on alternate days during flowering and fruiting. For trailing, spread dried twigs on the ground. Conduct weeding, and raking of the soil at the time of fertilizer application. Earthing up may be done during rainy season.

Pests

Fruit fly, epilachna beetle and red pumpkin beetle are important pests affecting ash gourd. They can be controlled as given under bitter gourd. Aphids can be controlled by spraying malathion 0.01 per cent or quinalphos 0.05 per cent.

Diseases

Powdery mildew, downy mildew and mosaic are important diseases in ash gourd. They can be controlled as in bitter gourd.

Harvesting can be done only after 10 days (at least) of insecticide/fungicide application. The fruit should be washed thoroughly in water before cooking.

[Top](#) [Back to Contents](#)

SOLANACEOUS VEGETABLES

Brinjal, chilli and tomato are the important solanaceous fruit vegetables grown in the state. The cultural operations of the above three crops are similar with only slight variations.

BRINJAL (*Solanum melongena*)

Varieties

Surya, Swetha and Haritha (bacterial wilt resistant open pollinated varieties), Neelima (bacterial wilt resistant F1hybrid), Pusa Purple Cluster.

Seed rate : 370-500 g ha⁻¹

Raising seedlings

Brinjal is a transplanted vegetable. Seeds are sown in the nursery and one month old seedlings are transplanted to the main field. For sowing the seeds, raised seed beds of 90 to 100 cm width and convenient length are prepared in open space with fertile topsoil to which well decomposed organic matter has been incorporated. After sowing the seeds, mulch with green leaves and irrigate with a rose-can daily in the morning. Remove the mulch immediately after germination of the seeds. Restrict irrigation one week before transplanting and irrigate heavily on the previous day of transplanting.

Time of planting

For rainfed crop, transplant the seedlings during May-June before the onset of southwest monsoon. Planting can also be done during September-October for irrigated crop.

Land preparation and transplanting

Land is prepared to a fine tilth by thorough ploughing or digging. Well rotten organic manure is incorporated in the soil and seedlings are transplanted in shallow trenches/pits during May or on ridges/levelled lands during rainy season. Transplanted seedlings may be given temporary shade for 3-4 days during summer.

Spacing

Transplant less spreading varieties like Swetha and Surya at 60 cm x 60 cm. For spreading varieties Haritha and Neelima, provide wider spacing of 75-90 cm x 60 cm.

Manuring

Apply well rotten FYM / compost @ 20-25 t ha⁻¹ at the time of land preparation and mix well with the soil. A fertilizer dose of 75:40:25 kg N:P₂O₅:K₂O per ha may be given. Half the dose of nitrogen, full phosphorus and half of potash may be applied as basal dose before transplanting. One fourth of nitrogen and half of potash may be applied 20-30 days after planting. The remaining quantities may be applied two months after planting. Application of 75:25:25 kg N:P₂O₅:K₂O per ha is optimum for getting maximum yield of fruits for the variety Swetha in the reclaimed alluvial soils of Kuttanad. However, the economic optimum dose was found to be 60:20:25 kg of N:P₂O₅:K₂O per ha.

Aftercultivation

Irrigate at three or four days interval during summer. Stake the plants if necessary. Weeding followed by fertilizer application and earthing up may be done at one and two months after transplanting.

Plant protection

For avoiding damping off of the seedlings in the nursery, sow the seeds as thin as possible in the raised beds prepared in the open area during summer months.

Follow mechanical removal and destruction of pest / disease affected portions for control of fruit and shoot borer and *Phomopsis* fruit rot. Spray carbaryl 0.15 per cent at an interval of 15-20 days to control fruit and shoot borer under large scale cultivation.

Uproot plants affected by little leaf and spray insecticides for further control. Cultivate resistant varieties like Surya, Swetha and Haritha and the hybrid Neelima in bacterial wilt prone areas.

For managing root knot nematode, nursery treatment with *Bacillus macerans* / *Paecilomyces lilacinus* @ 25 g/m² + drenching with the same @ 3 per cent solution 7 days after sowing can be recommended.

In general, insecticides of plant origin may be used, as far as possible.

[Top](#) [Back to Contents](#)

CHILLI (*Capsicum annuum*)

Varieties

High yielding varieties: Jwalasakhi, Jwalamukhi, Jwala, Pant C-1, K-2, Vellayani Athulya and Vellayani Samrudhi (Tolerant to shade and recommended for southern zone of Kerala.)

Bacterial wilt resistant varieties: Ujwala, Anugraha.

Seed rate: 1.0 kg ha⁻¹

Raising seedlings

Chilli is a transplanted crop. Seeds are sown in the nursery and one month old seedlings are transplanted to the main field. For sowing the seeds, raised seed beds of 90 to 100 cm width and of convenient length are prepared to which well decomposed organic matter has been incorporated. After sowing the seeds, mulch with green leaves and irrigate with a rosecan daily in the morning. Remove the mulch immediately after germination of the seeds. Restrict irrigation one week before transplanting and irrigate heavily on the previous day of transplanting.

Time of planting

For a rainfed crop, transplant the seedlings during May-June before the onset of southwest monsoon. Planting can also be done during September-October for an irrigated crop.

Land preparation and transplanting

Land is prepared to a fine tilth by thorough ploughing / digging. Well rotten organic manure is incorporated in the soil and seedlings are transplanted in shallow trenches / pits during May or on ridges / level lands during rainy season. Transplanted seedlings may also be given temporary shade for three to four days during summer.

Spacing

Transplant less spreading varieties at 45 cm x 45 cm. For spreading cultivars like White Kanthari provide a wider spacing of 75 cm x 45-60 cm.

Manuring

Apply well rotten FYM / compost @ 20-25 t ha⁻¹ at the time of land preparation and mix well with the soil. A fertilizer dose of 75:40:25 kg N:P₂O₅:K₂O per ha may be given. Half of nitrogen, full phosphorus and half of potash may be applied as basal dose before transplanting. One fourth of nitrogen and half of potash may be applied 20-30 days after planting. The remaining quantity may be applied two months after planting.

Aftercultivation

Irrigate at three to four days interval during summer. Stake the plants if necessary. Weeding followed by fertilizer application and earthing up may be done at one and two months after transplanting.

Plant protection

For avoiding damping off of the seedlings in the nursery, sow the seeds as thin as possible in raised beds prepared in the open area during summer months. Spray nursery and main field with 1 per cent Bordeaux mixture at monthly intervals during rainy season. Uproot and destroy the plants affected by bacterial wilt and mosaic.

Cultivate resistant varieties like Ujwala and Anugraha in bacterial wilt prone areas.

Spray quinalphos 0.05 per cent for control of mealy bugs and lace wing bugs. Dimethoate at 0.05 per cent is effective for controlling mites, aphids and other sucking insects.

Chilli mite (*Polyphagotarsonemus latus*)

It is an important sucking pest infesting all stages of plant. Feeding of mite causes downward curling of the leaves and become brittle and tubular. It can be managed by the application of neem oil 5 per cent and neem oil+garlic emulsion 2 per cent.

[Top](#) [Back to Contents](#)

TOMATO (*Solanum lycopersicum*)
Varieties

Bacterial wilt resistant varieties: Sakthi, Mukthi, Anagha and Vellayani Vijai.

High yielding variety: Pusa Ruby

Seed rate : 400 g ha⁻¹

Raising seedlings

Tomato is a transplanted vegetable. Seeds are sown in the nursery and one month old seedlings are transplanted to the main field. For sowing the seeds, raised seed beds of 90 to 100 cm width and of convenient length are prepared to which well decomposed organic matter has been incorporated. After sowing the seeds, mulch with green leaves and irrigate with a rose-can daily in the morning. Remove the mulch immediately after germination of the seeds. Restrict irrigation one week before transplanting and irrigate heavily on the previous day of transplanting.

Time of planting

Transplant the seedlings during October-November for an irrigated crop.

Land preparation and transplanting

Land is prepared to a fine tilth by thorough ploughing or digging. Well rotten organic manure is incorporated in the soil and seedlings are transplanted in shallow trenches / pits / levelled lands. Transplanted seedlings may be given temporary shade for three to four days during hot days.

Spacing

Transplant the seedlings at 60 cm x 60 cm.

Apply well rotten FYM/compost @ 20-25 t ha⁻¹ at the time of land preparation and mix well with the soil. A fertilizer dose of 75:40:25 kg N:P₂O₅:K₂O per ha may be given. Half the dose of nitrogen, full phosphorus and half of potash may be applied as basal before transplanting. One fourth of nitrogen and half of potash may be applied 20-30 days after planting. The remaining quantity may be applied two months after planting.

Aftercultivation

Irrigate at two or three days interval. Stake the plants if necessary. Weeding followed by fertilizer application and earthing up may be done at one and two months after transplanting.

Plant protection

For avoiding damping off of the seedlings in the nursery, sow the seeds as thin as possible in raised beds prepared in the open area. Spray nursery and main field with 1 per cent Bordeaux mixture at monthly intervals. Uproot and destroy the plants affected by bacterial wilt and mosaic. Cultivate resistant varieties like Sakthi, Mukthi and Anagha in bacterial wilt prone areas.

[Top](#) [Back to Contents](#)

COOL SEASON VEGETABLES

(*Ad hoc recommendation*)

CABBAGE (*Brassica oleracea* var. *capitata*)

Cabbage can be grown in high ranges during winter season. Well-drained sandy loam to clay loam soil is suited for this crop.

Varieties

September, Pusa Drum Head, Golden Acre, Kaveri, Ganga, Sri Ganesh and Pride of India.

Planting requirements

Since it is a cool season crop, sowing is done from August-November. Seed rate is 500-750 g ha⁻¹. Seeds are to be sown in nursery beds. Three to five weeks old seedlings are used for transplanting. Field is prepared by three or four ploughings. Seedlings are transplanted at a spacing of 45 cm x 45 cm.

Manures and fertilizers

Apply 25 t ha⁻¹ FYM or compost. Fertilizer dose is N:P₂O₅:K₂O 150:100: 125 kg ha⁻¹. Apply full dose of P₂O₅ and half dose of N and K₂O before transplanting. Apply remaining half dose one month after transplanting.

Aftercultivation

A continuous supply of moisture is necessary for proper development of heads. Very shallow hoeing should be done to remove weeds and to make the soil better aerated. In order to produce large heads, earth up plants one month after transplanting.

[Top](#) [Back to Contents](#)

CAULIFLOWER (*Brassica oleracea* var. *botrytis*)

Cauliflower can be grown during winter in high ranges. Well-drained sandy loam to clay loam soils are suited for the crop.

Varieties

Pusa Early Synthetic, Pusa Deepali, Early Patna, 74-6-C

Planting requirements

Since it is a cool season crop, sowing is to be done from Aug-Nov. Seed rate is 600-750 g ha⁻¹. Seeds are to be sown in nursery beds. Three to five week old seedlings are used for transplanting. Field is prepared by three or four ploughings. Seedlings are transplanted at a spacing of 60 cm x 45 cm.

Manures and fertilizers

Apply FYM or compost @ 25 t ha⁻¹ and fertilizers @ 150:100:125 N:P₂O₅:K₂O kg ha⁻¹. Apply full dose of P₂O₅ and half dose of N and K₂O before transplanting and remaining N and K one month after transplanting.

Aftercultivation

A continuous supply of moisture is necessary for proper development of curds. Very shallow hoeing should be done to remove the weeds and to loosen the soil for better aeration. In order to produce large curds, earth up the plant one month after transplanting.

[Top](#) [Back to Contents](#)

CARROT (*Daucus carota*)

Carrot can be grown in high ranges from August to January. Well-drained sandy loam soil is best suited for the crop.

Varieties

Pusa Kesar, Nantes, Pusa Meghali

Planting requirements

Seed rate is 5-6 kg ha⁻¹. It is usually sown on ridges to facilitate good root production. Ridges of about 20 cm height are made 45 cm apart and seeds sown 10 cm apart on the rows. The seed is mixed with fine sand and sown in rows by hand and covered with soil to make it firm around it.

Manures and fertilizers

Apply 25 t ha⁻¹ FYM before sowing and a fertilizer dose of 37.5 kg N, 62.5 kg P₂O₅ and 50 kg K₂O per ha as basal. Topdressing with 37.5 kg N per ha may be done one month after sowing.

Aftercultivation

It is necessary that enough soil moisture is available to help uniform seed germination and growth of plant. Uproot excess seedlings (thinning) three weeks after sowing leaving a plant to plant spacing of 10 cm to facilitate better tuber growth. Weeding should be done at regular intervals to keep down the weeds. Shallow hoeing is necessary to facilitate root growth. When the root starts growing, earthing up should be done.

[Top](#) [Back to Contents](#)

BEET ROOT (*Beta vulgaris*)

Beet root can be grown in high ranges from August to January. Well-drained sandy loam soils are best suited for the crop.

Varieties: Detroit Dark Red

Planting requirements

Seed rate is 7 to 8 kg ha⁻¹. It is usually grown on ridges to facilitate good root production. Ridges of about 20 cm height are formed 45 cm apart and seeds sown 15-20 cm apart on the rows. The seeds are mixed with fine sand and placed in rows by hand and covered with soil to make it firm around it.

Manures and fertilizers

Apply FYM 20 t ha⁻¹ as basal. N:P₂O₅: K₂O 75:37.5:37.5 kg ha⁻¹ is recommended. Full dose of P₂O₅ and K₂O and half dose of N are applied as basal. Remaining half dose of nitrogen is applied as topdressing when the plant starts growing vigorously.

Aftercultivation

It is necessary that enough soil moisture is available to help uniform seed germination and growth of plant. Thinning the population may be done as in carrot. Weeding should be done at regular intervals to keep down the weeds. Shallow hoeing is necessary to facilitate root growth. When the root starts growing, earthing up should be done.

[Top](#) [Back to Contents](#)

RADISH (*Raphanus sativus*)

Radish can be grown in high ranges from June to January. Well drained sandy loam soils are best suited for the crop.

Varieties

Japanese White, Arka Nishanth, Pusa Chethki, Pusa Reshmi and Pusa Desi.

Planting requirement

Seed rate is 7 to 8 kg ha⁻¹. It is usually grown on ridges to facilitate good root production. Ridges of about 20 cm height are taken 45 cm apart and plants are grown 10 cm apart on the rows. The seed is mixed with fine sand and sown in rows by hand, covered with soil to make it firm around it.

Manures and fertilizers

Apply 20 t ha⁻¹ FYM as basal. N: P₂O₅: K₂O 75:37.5:37.5 kg ha⁻¹ is the fertilizer requirement. Full dose of P₂O₅ and K₂O and half dose of N are applied as basal. Remaining half dose of nitrogen is applied as topdressing when the plant starts growing vigorously.

Aftercultivation

It is necessary that enough soil moisture is available to help uniform seed germination and growth of plant. Thinning may be done at 10 cm distance as in carrot. Weeding should be done at regular intervals to keep down weeds. Shallow hoeing is necessary to facilitate root growth. When the roots start growing, earthing up should be done

[Top](#) [Back to Contents](#)

POTATO (*Solanum tuberosum*)

Potato can be successfully cultivated in the high ranges of Kerala. It is being cultivated in the rain shadow areas of Idukki district throughout the year. A day temperature of 20-30°C is optimum for growth and tuberisation in potato. Tuber formation is adversely affected, if the temperature goes above 30°C.

Season

Crops can be raised as shown below in the eastern part of Idukki district.

Summer : March-April

Autumn : August-December

Spring : January-February

Varieties

Among the high yielding varieties, Kufri Jyothi (early), Kufri Muthu (medium), and Kufri Dewa (late) can be tried in the high ranges of Kerala.

Soil

Loose friable sandy loam or silt loam, rich in organic matter are ideal for potato. Hard clay should be avoided. Optimum pH range is 5.2-7.0

Planting

Whole potato tuber or cut pieces (50-60 g size) longitudinally cut from bud-end to stem end can be used for planting. For planting 1 ha, 1000-2000 kg seed tubers are required. Seed tubers are treated with 1 ppm of GA 3 for one hour and then dried in shade for getting uniform sprouting. Tubers are filled in gunny bags after drying and kept in vertical position in well-ventilated dark room for 10 days for encouraging sprouting. Seed pieces should be treated with mancozeb (@ 1 kg in 450 litres of water) before planting to protect them from soil borne diseases. Tubers can be planted on ridges 50-60 cm wide at a spacing of 15-20 cm between the plants. Earthing up is needed during the growing phase (30 days after planting) and 70 days after planting.

Manuring

A basal application of FYM (20 t ha⁻¹) is required during field preparation. Apply 60 kg N, 100 kg P₂O₅ and 120 kg K₂O as basal. Top dressing with 60 kg N, 30 days after planting at the time of first earthing up is essential.

Plant protection

Early blight and late blight are the important fungal diseases. Spraying zineb (2 g per litre of water) is effective to control early blight. Copper fungicides can control late blight. Cut worms, aphids and jassids are common pests of potato. Dusting with carbaryl 10 per cent DP immediately after planting can control cut worms. Spraying carbaryl 0.2 per cent controls leaf eating caterpillars. To control aphids and jassids spraying dimethoate (0.05 per cent) is effective.

[Top](#) [Back to Contents](#)

GARLIC (*Allium sativum*)

Garlic requires cool and moist period during vegetative growth and a dry spell during maturity of the bulbs. Fertile, well-drained loamy soils are ideal for garlic cultivation. Heavy clay soils may result in deformed bulbs. In high ranges of Kerala garlic can be planted during October- November.

Varieties: Ooty-1, G 50

Planting requirements

Cloves or bulbils are used for propagation. For planting one hectare, 500 kg of cloves is required. The cloves for planting should be stored for 2-3 months after harvest and cloves weighing 4 g are ideal for planting. The cloves should be soaked in water followed by dipping for 15 minutes in a solution containing 1 g of carbendazim dissolved in 1 litre of water for 15 minutes. After drying in shade, cloves can be used for planting. Dig the land thoroughly and prepare beds of 15 cm height at a width of 1 m and of

convenient length. The cloves should be dibbled at a spacing of 15 cm x 8 cm. Germination will start on the fifth day and it will be completed within 10-15 days.

Manuring

Apply N:P₂O₅: K₂O @ 60:120:120 kg ha⁻¹ 20 days after transplanting (DAP). Top dressing of N should be done @ 60 kg ha⁻¹, 45 DAP.

Aftercultivation

Earthing up should be done 60 DAP

Plant protection

To control blast, spray mancozeb (2 g l⁻¹)

Harvest

Harvesting can be done 120-130 DAP. Yield may vary from 5-10 t ha⁻¹.

[Top](#) [Back to Contents](#)

MINOR VEGETABLES

Dolichos bean (*Lablab purpureus*)

Pole varieties : Hima, Grace and Pusa Early Prolific

Bush variety : Arka Vijay

Pole varieties are sown in pits (three plants per pit) at a spacing of 1.25 m x 0.75 m and bush varieties in ridges and furrows at a spacing of 60 cm x 15 cm. Seeds are to be sown during July-August. The plants are trailed over pandals, trellis or stakes. FYM is applied @ 20 t ha⁻¹. N:P₂O₅: K₂O recommendation for the crop is 50:100:50 kg ha⁻¹. The leaf caterpillar is a common pest of the crop. It is also affected by *Fusarium* wilt, collar rot, anthracnose and powdery mildew. Average yield is 6-10 t ha⁻¹.

Winged bean

(*Psophocarpus tetragonolobus*)

The common varieties in use are Revathy, PT-62, PT-16, PT-49 and PT-2. Seeds @ 15 to 20 kg ha⁻¹ are planted at a spacing of 125 cm x 50 cm during August-September and are trailed over pandal, trellis or stakes. FYM is applied @ 20 t ha⁻¹. N, P₂O₅ and K₂O recommendation for the crop is 50:100:50 kg ha⁻¹.

The crop is comparatively free from pests and diseases. Average yield is 10-15 t ha⁻¹.

Cluster bean (*Cyamopsis tetragonoloba*)

Pusa Naubahar and Pusa Sadabahar are the common varieties. Seeds @ 10 to 12 kg ha⁻¹ are planted at a spacing of 45-60 cm x 20-30 cm in February-March and June-July. During rainy season, the seeds are sown 2-3 cm deep on ridges and in furrows during summer months. FYM is applied @ 25 t ha⁻¹. N, P₂O₅ and K₂O recommendation for the crop is 20:60:80 kg ha⁻¹. Aphids and powdery mildew are the common pest and disease of the crop. Average yield is 5 to 6 t ha⁻¹.

Sword bean (*Canavalia gladiata*) and Jack bean (*Canavalia ensiformis*)

Sword bean is trailing and red seeded while Jack bean is bushy and white seeded. Sword bean is to be planted at a spacing of 4 m x 3m whereas Jack bean is to be planted at 60 cm x 60 cm.

May-June and September-October are the usual sowing time and the seed rate followed is one or two seeds per pit. FYM is applied at the rate of 5 t ha⁻¹. The N:P₂O₅:K₂O mixture (7:10:5) may be applied as basal dose and top dressing at several splits. There is no serious pest or disease incidence in the crop. Average yield is 10-15 kg per plant.

Clove bean (*Ipomoea muricata*)

The crop can be grown throughout the year and are trailed over trellis or stakes. The seeds @ 6-7 kg ha⁻¹ are planted at a spacing of 1.0 m x 0.6 m. FYM is applied @ 10 t ha⁻¹. N:P₂O₅:K₂O recommendation for the crop is 35:50:25 kg ha⁻¹. There is no serious pest or disease incidence in the crop. Average yield is 5-6 t ha⁻¹.

Ivy gourd (*Coccinia grandis*)

Variety - Sulabha

Local varieties are grown in May-June and September-October by trailing over pandals and stakes. Stem cuttings with three or four nodes and 30-40 cm length, selected from high yielding female vines are used as planting material. These are planted at a spacing of 4 m x 3 m. FYM @ 25 kg per pit is given in two doses. No serious pests or diseases are reported except mild attack of fruit flies and gall insects.

Smooth gourd (*Luffa cylindrica*)

Pusa Chickni is the common variety in use. The crop is planted in February-March and May-June at a spacing of 2 m x 2 m. The seed rate is 2.5-3 kg ha⁻¹. The crop is trailed over pandal, stakes or trellis. FYM @ 25 t ha⁻¹ is given in two doses. N:P₂O₅:K₂O recommended for the crop is 70:25:25 kg ha⁻¹. No serious pests or diseases are reported. The average yield is 10-15 t ha⁻¹.

Ridge gourd (*Luffa acutangula*)

The important varieties are Haritham, Pusa Nasdhar Co-2 and Deepthi. The crop is usually sown during February-March and May-June. Seed rate recommended is 2.5-3.0 kg ha⁻¹ with a spacing of 2 m x 2 m. FYM @ 25 t ha⁻¹ and N:P₂O₅:K₂O @ 70:25:25 kg ha⁻¹ are recommended. It is usually trailed over pandals or trellis. Average yield is 10 to 15 t ha⁻¹.

Bell pepper

(*Capsicum annuum* var. *grossum*)

Hungarian Wax, California Wonder and Early Calwonder are the promising varieties. Plant the crop during September-October at a spacing of 60 cm x 30 cm. The seed rate is 400-600 g ha⁻¹. Raising of seedlings, transplanting, irrigation etc. are same as in chilli. FYM @ 25 t ha⁻¹ and N:P₂O₅:K₂O @ 150:75:50 kg ha⁻¹ are to be given. The average yield is 12-15 t ha⁻¹.

Drumstick (*Moringa oleifera*)

Variety: Anupama (Early flowering habit and suitable for Central Zone.) The major planting season is May-June. Stem cuttings of 1.0-1.5 m length and 15-20 cm girth are used as planting material. Plant the cuttings in polybags and later sprouted cuttings can be shifted to main field. For one hectare, 625 cuttings are required. These are planted at a spacing of 4 m x 4 m. FYM @ 10-20 kg per pit and N: P₂O₅:K₂O @ 60:80:40 g per pit are recommended. Green caterpillar and hairy caterpillar are the common pests. The average yield is 10-15 kg per tree per year.

Chekkurmanis (*Sauropus androgynus*)

Stem cuttings of 6-12 months old, 20-30 cm length are to be planted in May-June. These are usually grown on borders of kitchen gardens. To check the height of the plant and to get frequent harvests, the tips are clipped off intermittently. FYM @ 5 kg per plant per year and N:P₂O₅:K₂O (7:10:5) mixture @ 30 g per plant are recommended. The average yield is 2 _ 5 kg per m² per year.

Indian spinach (*Basella* sp.)

Seeds or stem cuttings of 20-30 cm length are to be planted during May-June and September-October. The spacing recommended is 1.0 m x 0.6 m. These are usually trailed over pandals or stakes. FYM @ 2-5 kg per m² is to be given. The average yield is 1-2.5 kg per m².

Water leaf (*Talinum triangulare*)

This is a shade loving leafy vegetable grown in May-June and September-October. Semi hard stem cuttings of 10-15 cm length are planted at a spacing of 30 cm x 10 cm. FYM @ 2-5 kg per m² is given. The average yield is 2.0-2.5 kg per m².

Curry leaf (*Murraya koenigii*)

It is usually planted in May-June. Root suckers are used as planting material. The recommended spacing is 4 m x 4 m with 625 plants per hectare. FYM @ 10 kg per plant per year is given. N:P₂O₅:K₂O @ 60:80:40 g per adult plant per year is recommended. Major pests are citrus butterfly and psyllid. *Diaphorina* pink disease is also seen. The average yield is 2-2.5 kg per m².

Table 25. Waiting periods for insecticides on vegetables

Vegetable	Carbaryl	Quinalphos	Malathion	Dimethoate
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Okra	5 days	3 days	3 days	3 days
Bitter gourd	11 days	5 days	4 days	2 days
Brinjal	5 days	3 days	3 days	3 days
Snake gourd	5 days	6 days	1 day	3 days
Tomato	6 days	-	1 day	5 days
Chilli	5 days	-	1 day	4 days

Note: Washing vegetables in 2% table salt solution or 2% vinegar and thorough washing in water using scrubber were found to remove residues of contact insecticides.

[Top](#) [Back to Contents](#)

NON-CHEMICAL CONTROL OF PESTS OF VEGETABLES

(Ad hoc recommendations)

The American serpentine leaf miner is a recently introduced polyphagous pest infesting crops like cucurbits, brinjal, cowpea, sesame, groundnut and ornamentals. Larvae mine into the leaves and exhibit irregular serpentine like lines. As a result of the attack, leaves dry up and cause extensive damage. Spray neem oil emulsion for controlling the pest. Amaranth leaf webber and grasshopper, okra leaf roller, epilachna beetle on brinjal and bitter gourd, aphids, jassids and mealy bugs on brinjal and okra can be controlled by 4 per cent leaf extracts of neem/ thevetia / clerodendron with soap water. Okra fruits can be protected from infestation by fruit borers by spraying 4 per cent leaf extracts of thevetia / neem.

Preparation of plant extract emulsion

Soak 400 g of leaf powder (leaves dried under shade and powdered) in one litre of water for 24 hours and filter through muslin cloth. Dissolve 400 g ordinary bar soap shavings in 9 litres water. Pour this soap solution to the plant extract and mix thoroughly. This forms 4 per cent emulsion of plant extract.

[Top](#) [Back to Contents](#)

TIPS FOR VEGETABLE SEED PRODUCTION

General principles

The seed production programme envisages to produce genetically pure quality seeds and to store them in a viable condition for a reasonable period of time, until it reaches the farmers. The seeds should have genetic purity, uniformity in size and shape, high germination and vigour. The seeds should be free from mechanical damages, insect and fungal infestation and other crop and weed seeds. A commercial seed production programme has three aspects - seed production, seed processing and seed storage.

a. Seed production

The following aspects are important in this:

1. Basic knowledge on the specific requirement of the crop (climate, soil requirement etc.), specific characteristics of the variety, pests and diseases and their control measures are essential before taking up the seed production programme.
2. In general, September to January is the most suitable season for taking up seed production in Kerala.
3. Site selected for seed production should be open, receiving good sunlight, well drained and fertile soil, free from infectious pest and disease organisms.
4. Seeds for multiplication should be obtained from reliable sources.
5. Proper isolation distance should be maintained between varieties and related species.
6. Scientific roguing (removal of off-types at nursery stage, vegetative phase, flowering, fruiting and harvest stages) should be practised.
7. All plants infected by diseases should be removed from the seed production plot. No objectionable weeds are permitted in seed production plot.
8. In general, for most vegetable crops, taking one or two vegetable harvests is found ideal for economic seed production.
9. Provide one additional topdressing with N and K₂O at fruit development phase, adequate irrigation and plant protection measures.

10. General cultivation and plant protection practices recommended for vegetable production can be adopted in seed crop also.

11. Harvest the crop at optimum fruit maturity, since immature and over mature fruits affect the seed quality.

b. Seed processing

1. Seed processing involves extraction of seeds from the fruits and reducing the seed moisture content to a level of 6-8 per cent.

2. Wet and dry methods of seed extraction are adopted depending on the nature of the crop. Slow drying at low temperature (below 38°C) is advisable rather than quick drying at high temperature.

3. Exposing seeds to open sunlight during peak hours of sunshine (12 noon to 3 p.m.) should be avoided.

4. Spread the seeds in thin layer and give frequent raking for aeration while drying to avoid fungal infestation.

5. Clean the seeds by removing inert matter, damaged and underdeveloped seeds etc. to get uniform quality seeds.

c. Seed storage

1. Store seeds under cool and dry conditions to maintain the viability

2. Seed moisture content of 6-8 per cent, atmospheric temperature of 22°C and relative humidity of 45 per cent are the most ideal conditions for seed storage. The sum of storage temperature (°C) and humidity (per cent) should not exceed 80.

3. Pre-storage seed treatment with fungi-cides (captan or thiram @ 2.5 g kg⁻¹ of seeds) and insecticides (carbaryl 10 per cent DP @ 10 g kg⁻¹) would protect the seeds from various fungal and insect infestations.

4. Store the treated seeds in 700 gauge polythene bags in sealed conditions. Specific recommendations for seed production of the individual crops other than the general aspects mentioned above are given below:

Solanaceous vegetables

Tomato

The best time for planting tomato for seed production is October. Give an isolation distance of 50 m for foundation seed (FS) production and 25 m for certified seed (CS) between varieties. Rogue out off-types and virus infected plants. Maximum off-types and diseased plants permitted is 0.1 per cent each only. Seeds should have a minimum purity of 98 per cent and germination of 70 per cent. Maximum permitted inert matter content is 2 per cent, other crop seeds 0.1 per cent, weed seeds 0.1 per cent, and maximum moisture content 8 per cent.

Brinjal

Give an isolation distance of 200 m for FS and 100m for CS. Remove off-types and plants infected by little leaf disease. No objectionable weeds are permitted in the seed production plot. Maximum level of off-types and plants infested by designated diseases are 0.1 per cent each. Seeds should have a minimum purity of 98 per cent and germination of 70 per cent. Maximum inert matter content permitted is 2 per cent, other crop seeds 0.1 per cent, weed seeds 0.1 per cent and maximum moisture content 8 per cent.

Chilli

Provide an isolation distance of 400 m for FS and 200 m for CS. Remove off-types and plants infected by virus diseases. No objectionable weeds are permitted in the seed production plot. Maximum level of off-types and plants infested by designated diseases are 0.1 per cent each. Fruits of 45-50 days maturity may be harvested for seed extraction in the case of Jwalasakhi and Ujwala. Seeds should have a minimum purity of 98 per cent and germination of 70 per cent; maximum inert matter content permitted is 2 per cent, other crop seeds 0.1 per cent, weed seeds 0.1 per cent and maximum moisture content 8 per cent.

Okra

Provide an isolation distance of 400 m for FS and 200 m for CS. Remove off-types and plants infected by yellow vein mosaic disease. No objectionable weeds are permitted in the seed production plot. Maximum level of off-types and plants infected by yellow vein mosaic is 0.1 per cent each. It is economical to take two vegetable harvests and then retain the crop for seed production. Fruits of 36 days maturity may be harvested for seed extraction in the case of Arka Anamika. At this stage the pod colour completely turns to brown and tips dry. Seeds can be stored in polythene bags of 700 gauge thickness. Seeds should have a minimum purity of 99 per cent and germination of 65 per cent. Maximum inert matter content permitted is 1 per cent; no seeds of other crops or weeds are permitted; and maximum moisture content permitted for open storage is 10 per cent and for storing in moisture proof containers is 8 per cent.

Cucurbits

General

An isolation distance of 800 m for FS and 400 m for CS is required between varieties and related species. Remove off-types, wild cucurbits and plants infected by designated diseases. No objectionable weeds are permitted in the seed production plot. Maximum level of off-types and plants infected by yellow vein mosaic diseases is 0.1 per cent each. Seeds should have a minimum purity of 99 per cent and germination of 60 per cent. Maximum inert matter content permitted is 1 per cent. No seeds of other crops or weeds are permitted and maximum moisture content should not exceed 7 per cent.

Bitter gourd

Taking two vegetable harvests and then leaving the crop for seed production is economical. Ripe fruits of 24 days maturity in Preethi, when the whole fruits turn to bright orange colour, can be harvested for seed extraction. Seeds may be dried in the sun, avoiding peak sunshine hours of 12 noon to 3 p.m. Seeds can be stored in 700 gauge thick polythene bags.

Snake gourd

Fruits can be harvested for seed extraction 36 days after anthesis (in TA 19), when yellowing of fruits start from the stylar end. Big and medium sized fruits (above 85 cm length and 2 kg weight in Kaumudi) give maximum quantity of quality seeds.

Oriental pickling melon

Harvest fully ripe fruits with deep orange colour (30 days after anthesis in Mudicode Local), when the vines wither. Select big and medium sized fruits (above 1.25 kg in Mudicode Local) for quality seeds.

Machine extraction (without fermentation of pulp) and drying under shade for one day and then in sun avoiding peak hours (12 noon to 3.00 pm) to 8 per cent moisture gives good quality seeds.

Ash gourd

Ash gourd fruits of 70 days maturity after anthesis are suitable for seed extraction. At this stage the vines wither and the sticky thick ashy coating on the fruits dries into white powder, which can be removed on rubbing. Fruits of medium and large size (above 5 kg in KAU Local) give bolder, quality seeds. It is advisable to have a post-harvest storage of fruits for three months to get higher germination.

Manual extraction of pulp from the fruits and fermenting the pulp for 48 hours is better to get quality seeds without mechanical damage.

Acid treatment of pulp using 2 per cent HCl (1:10) for 30 minutes followed by drying under shade to 8 per cent moisture also gives good quality seeds.

Pre-storage treatment of seeds with captan @ 2.5 g kg⁻¹ and storing in sealed polythene bags of 700 gauge thickness is the best for seed storage.

Water melon

Manual extraction with acid treatment using 1 per cent HCl (1:10) for 30 minutes and drying under shade for one day and then in sun, avoiding peak hours (12 noon to 3.00 p.m) to 8 per cent moisture gives good quality seeds.

Vegetable cowpea

Dried pods can be harvested for seed purpose. These pods are further dried in the sun before seed extraction. Good drying and pre-storage seed treatment of seeds (with carbaryl 10 per cent DP @ 10 g per kg seeds) is essential to protect them from *Bruchus* and fungal infestations.

Amaranth

It is economical to take one vegetable harvest at 30 days after planting and then leaving the crop for seed production.

[Top](#) [Back to Contents](#)

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